§2.7 (Cont.)

In exercises 5–16, find the derivative $y'(x)$ implicitly.

5. $x^2y^2 + 3y = 4x$

6. $3xy^3 - 4x = 10y^2$

7. $\sqrt{xy} - 4y^2 = 12$

8. $\sin xy = x^2 - 3$

9. $\dfrac{x+3}{y} = 4x + y^2$

10. $3x + y^3 - \dfrac{4y}{x+2} = 10x^2$

11. $\cos(x^2y) - \sin y = x$

12. $x \sec y - 3y \sin x = 1$

13. $y^2\sqrt{x+y} - 4x^2 = y$

14. $x \cos(x+y) - y^2 = 8$

15. $\tan(y^2+3) - xy^2 = 2x$

16. $y \cos x^2 - \dfrac{3}{\sqrt{y^2+2}} = x^2 + 1$

In exercises 17–22, find an equation of the tangent line at the given point. If you have a CAS that will graph implicit curves, sketch the curve and the tangent line.

17. $x^2 - 4y^3 = 0$ at $(2, 1)$

18. $x^2y^2 = 4x$ at $(1, 2)$

19. $x^2y^2 = 3y + 1$ at $(2, 1)$

20. $x^3y^2 = -2xy - 3$ at $(-1, -3)$

21. $x^4 = 4(x^2 - y^2)$ at $(1, \frac{\sqrt{3}}{2})$

22. $x^4 = 8(x^2 - y^2)$ at $(2, -\sqrt{2})$

In exercises 23–28, find the second derivative $y''(x)$.

23. $x^2y^2 + 3x - 4y = 5$

24. $x^{2/3} + y^{2/3} = 4$

25. $y^2 = x^3 - 6x + 4\cos y$

26. $3xy + 2y - 3x = \sin y$

27. $(y - 1)^2 = 3xy + \cos(4y)$

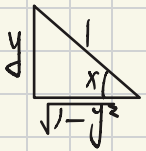
28. $(x + y)^2 - \sin(y + 1) = 3x$

32. Suppose that f is a differentiable function such that $f(\sin x) = x$ for all x. Show that, for $-1 < x < 1$, $f'(x) = \frac{1}{\sqrt{1-x^2}}$.

Let $y = \sin x$, then $\cos x = \sqrt{1-y^2}$

$\Rightarrow f'(\sin x) \cdot \cos x = 1$

$\Rightarrow f'(\sin x) = \frac{1}{\cos x}$



ค่อย ๆ หา answer

§2.8

In exercises 1–6, check the hypotheses of Rolle's Theorem and the Mean Value Theorem and find a value of c that makes the appropriate conclusion true. Illustrate the conclusion with a graph.

1. $f(x) = x^2 + 1, [-2, 2]$

2. $f(x) = x^2 + 1, [0, 2]$

3. $f(x) = x^3 + x^2, [0, 1]$

4. $f(x) = x^3 + x^2, [-1, 1]$

5. $f(x) = \sin x, [0, \pi/2]$

6. $f(x) = \sin x, [-\pi, 0]$

8. Prove that $x^3 + 4x - 3 = 0$ has exactly one solution.

25. Prove that $|\sin a| < |a|$ for all $a \neq 0$ and use the result to show that the only solution to the equation $\sin x = x$ is $x = 0$. What happens if you try to find all intersections with a graphing calculator?

§3.1

In exercises 1–6, find the linear approximation to $f(x)$ at $x = x_0$. Use the linear approximation to estimate the given number.

1. $f(x) = \sqrt{x}, x_0 = 1, \sqrt{1.2}$
2. $f(x) = (x+1)^{1/3}, x_0 = 0, \sqrt[3]{1.2}$
3. $f(x) = \sqrt{2x+9}, x_0 = 0, \sqrt{8.8}$
4. $f(x) = 2/x, x_0 = 1, \dfrac{2}{0.99}$
5. $f(x) = \sin 3x, x_0 = 0, \sin(0.3)$
6. $f(x) = \sin x, x_0 = \pi, \sin(3.0)$

49. Show that Newton's method applied to $x^2 - c = 0$ (where $c > 0$ is some constant) produces the iterative scheme $x_{n+1} = \frac{1}{2}(x_n + c/x_n)$ for approximating $\sqrt{c}$. This scheme has been known for over 2000 years. To understand why it works, suppose that your initial guess (x_0) for $\sqrt{c}$ is a little too small. How would c/x_0 compare to $\sqrt{c}$? Explain why the average of x_0 and c/x_0 would give a better approximation to $\sqrt{c}$.

50. Show that Newton's method applied to $x^n - c = 0$ (where n and c are positive constants) produces the iterative scheme $x_{n+1} = \frac{1}{n}[(n-1)x_n + cx_n^{1-n}]$ for approximating $\sqrt[n]{c}$.